DRINIX × CITI HACKATHON · PITCH DECK

TrustShield UPI

Smart Payment Warnings and Safe Cash Transit Framework

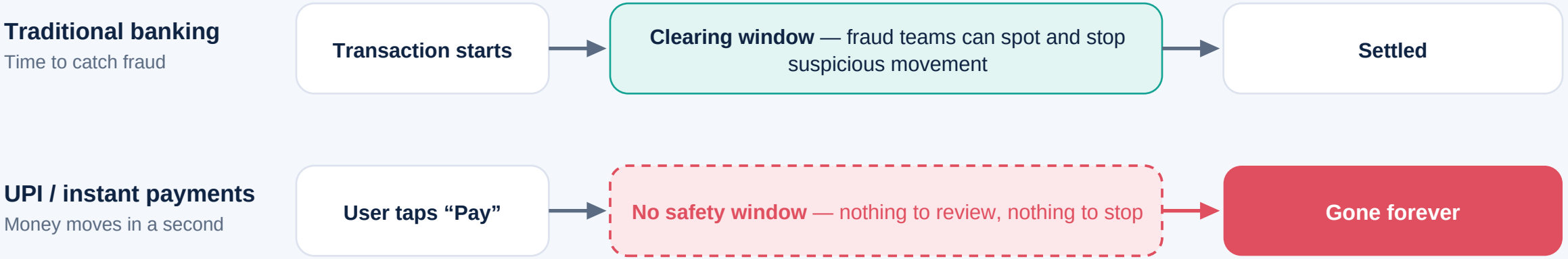
Fast payments that are genuinely safe — and inclusive for everyone, including visually impaired users.



Smart Alerts | VoicePay | Locked AutoPay | Secure Transit Pool

THE PROBLEM

Real-time payments removed the safety window



Urgency is the weapon

Fake hospital emergencies and “pay now or lose power” bills push victims to paste unknown payment IDs without thinking.

Generic warnings fail

Apps show the same pop-up for every transaction. Users get warning fatigue and tap “Proceed” just to clear the screen.

Instant finality hurts most

Once money leaves, it is gone. The structural gaps hit disabled users hardest — blind users cannot read warnings at all.

THE PROBLEM

Four gaps that leave everyday users exposed

1 First-time payments

- Scammers create false urgency
- Same generic pop-up every time → warning fatigue
- No way to tell a verified merchant from a private account posing as a business

2 Hidden AutoPay

- Confusing layouts and hidden settings enrol users silently
- Frequency, maximum amount and true receiver identity are buried
- Like signing a blank digital check

3 Trading with strangers

- Buyers fear paying upfront
- Holding money until delivery lets a dishonest buyer take the item and claim “not received”
- Honest seller loses cash and inventory

4 Inaccessible to the blind

- Cannot read warning pop-ups
- Cannot verify long UPI IDs or spot business flags
- No secure audio mechanism to govern payments and transit states

The common thread

Payments are **instant and final**, warnings are **generic**, and critical details are **hidden or unreadable** — so scammers win in the gap.

THE SOLUTION

TrustShield UPI: a smart security and accessibility layer

Real checkpoints matched to the exact situation — not generic warnings users learn to ignore. Payments stay fast; scams lose their advantage.

PILLAR 1

Smart First-Time Alerts

Detects brand-new payees, shows Verified Business vs Unverified Private Account, adds a 3-second pause and a scam-matched question.

Stops urgency scams

PILLAR 2

Locked-by-Default AutoPay

Every auto-debit starts OFF. One clear mandate card, two-step approval, 24-hour pre-charge alerts and one-click cancel.

Ends hidden subscriptions

PILLAR 3

Secure Transit Pool

Money for a stranger deal waits in a bank-managed pool and is released only when the seller scans the buyer's one-time code.

Protects buyer and seller

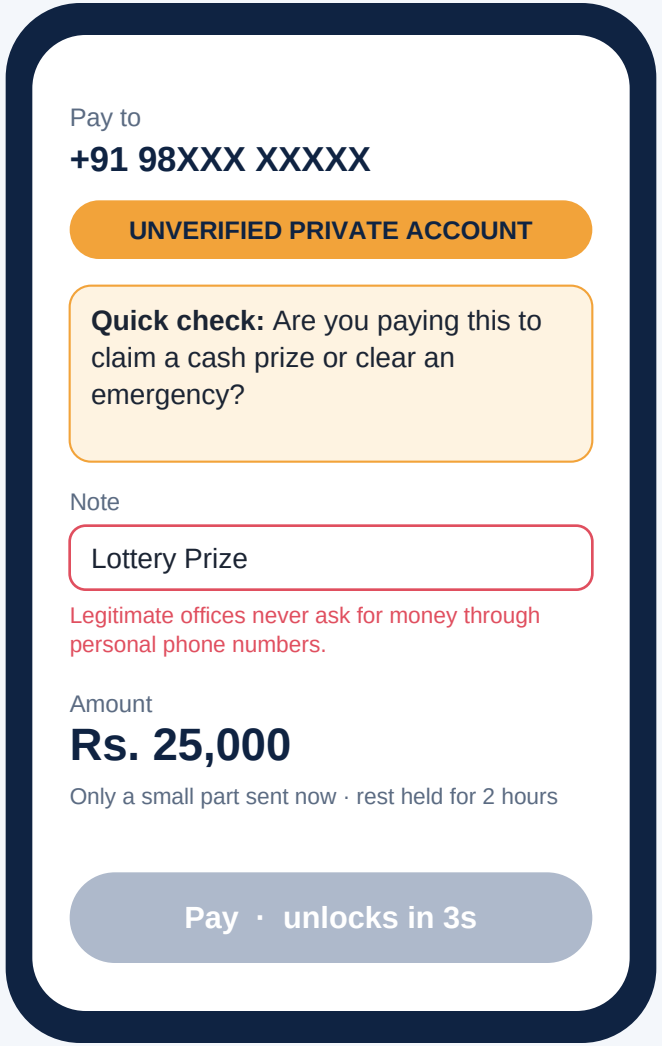


VoicePay — voice-first independence across all three pillars

Voice commands to pay · spoken recipient, amount and risk warnings · mandate details read aloud before approval · delivery PIN shared privately · payment status and history by voice query.

PILLAR 1

Smart first-time alerts that break the scammer’s rush



Trigger: new phone number or UPI ID detected → app switches to high-alert mode.

Business vs private check

The destination account is checked instantly. A bright banner shows “Verified Business” or “Unverified Private Account” so a personal number cannot pose as an official business.

Smart safety friction

For risky numbers the Pay button is locked behind a 3-second countdown, alongside a simple question that mirrors common scams.

Payment note scanning

Notes like “Customs Duty Fee”, “Lottery Prize” or “Electricity Bill Penalty” sent to a personal number trigger a clear alert (read aloud by VoicePay).

Safety hold on large sums

For unverified first-time transfers only a tiny fraction goes instantly; the rest is held for two hours so the victim can cancel in time.

ACCESSIBILITY

VoicePay: full payment independence for visually impaired users

VoicePay conversation

Illustrative example

“Pay Ravi 500 rupees.”

I found two contacts named Ravi: **Ravi Kumar**, ending 4321, and **Ravi S**, ending 9876. Which one?

“Ravi Kumar.”

Paying **Ravi Kumar**, 500 rupees. This is a **first-time** payee on an **unverified private account**. Say “confirm” to continue or “cancel” to stop.

“Confirm.”

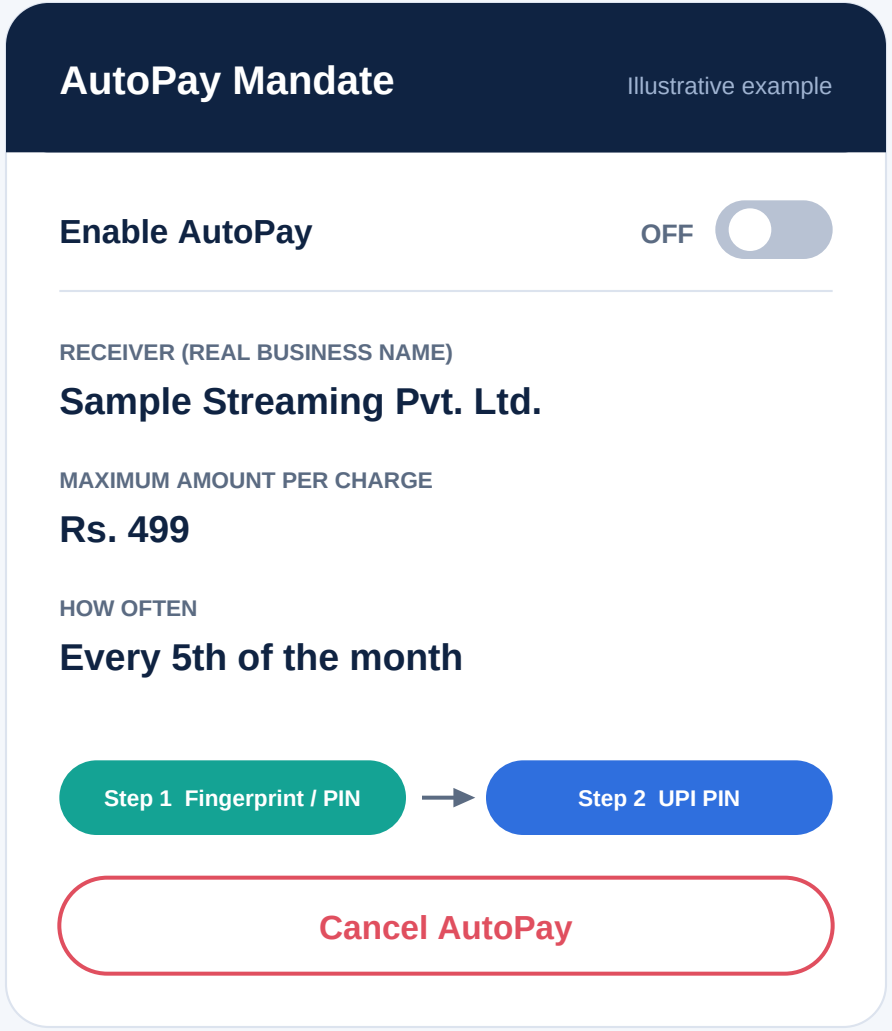
Payment sent. 500 rupees to Ravi Kumar.

1. **Speak the payment**
   Say the recipient’s name or UPI ID along with the amount — no screen needed.
2. **Hear it read back**
   The app reads the recipient and amount aloud and asks for confirmation. Same-name contacts are told apart by distinguishing details.
3. **Hear the risk**
   Fraud-risk warnings and business flags are spoken, and extra verification is requested when necessary.
4. **Authorised payment completes**
   After the authorised payment process finishes, the app announces the transaction status.
5. **Ask your history**
   Payment history is fully accessible through simple voice queries.

Voice runs through every pillar: note alerts read aloud, mandate details spoken before approval, AutoPay paused by voice, delivery PIN shared privately.

PILLAR 2

Locked-by-default AutoPay with full transparency



1 Completely locked by default

Auto-debit is OFF out of the box. Apps and websites cannot pre-check boxes or use tricky layouts — the user must switch it ON.

2 One simple mandate card

Before anything goes live, the hidden details come together: the receiver's real name, the maximum charge, and how often it is taken. VoicePay reads them aloud.

3 Two-step approval

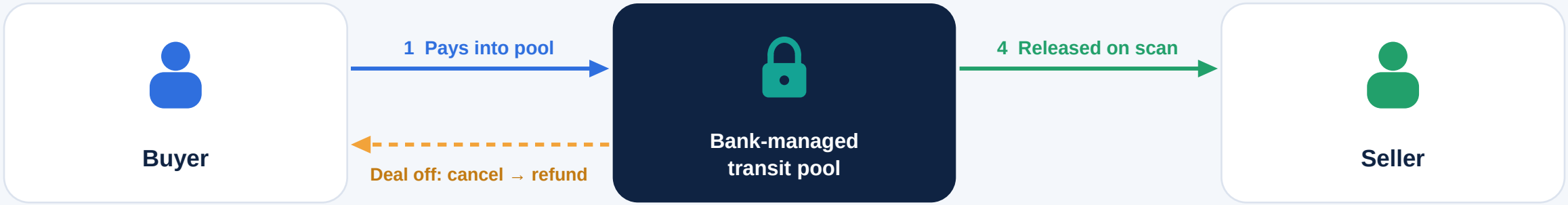
Lock 1: fingerprint or app-lock PIN proves the user holds the phone. Lock 2: the official secure page asks for the UPI PIN to finalise.

4 One-click cancel and early alerts

A dashboard lists every active AutoPay with a large Cancel button. A notice 24 hours before each charge lets users pause by tap or voice.

PILLAR 3

Secure transit pool: trust between strangers, both ways



2 One-time code created

The moment funds enter the pool, the buyer's app creates a one-time QR code and a 4-digit delivery PIN (VoicePay shares it privately).

3 Safe physical handover

At the meetup the buyer inspects the item. Only once happy do they show the QR or share the PIN, and the seller scans it.

4 Instant, permanent payment

The moment the seller's app matches the code, funds go from the pool to the seller's bank account.

Protects the buyer

- Money never goes straight to an unknown seller
- Item is inspected before the code is shared
- If the deal falls through, cancel and the money returns to the bank account

Protects the seller

- Funds are already secured in the pool — not just a promise
- Permanent digital proof the code was scanned
- Buyer cannot later claim "item never received" to force a refund

THE DIFFERENCE

Today’s payment apps vs TrustShield UPI

Scenario	Today	With TrustShield UPI
First-time transfer	✗ The same generic pop-up for every payment; users tap “Proceed” out of habit.	✓ Situation-specific alert, 3-second pause, scam-matched question, note scanning and a 2-hour hold on large sums.
Business or private?	✗ A private account can pose as a business with no clear signal.	✓ Banner on every first payment: “Verified Business” or “Unverified Private Account”.
AutoPay sign-up	✗ Hidden settings; frequency, maximum amount and receiver identity are buried.	✓ OFF by default, one mandate card, fingerprint/PIN plus UPI PIN, 24-hour pre-charge alert, one-click cancel.
Deal with a stranger	✗ Buyer risks paying upfront — or a held payment lets a dishonest buyer claim “not received”.	✓ Transit pool with one-time QR/PIN: seller paid on scan with permanent proof; buyer refunded if cancelled.
Visually impaired user	✗ Cannot read pop-ups, verify long UPI IDs or spot business flags.	✓ VoicePay: voice commands, spoken warnings and confirmations, spoken status and history.

CONCLUSION

A digital wallet that actively protects you

TrustShield UPI keeps the incredible speed of real-time banking while creating an environment where fraud cannot thrive.



Fraud stopped early

Smart checkpoints, friction and holds act before the money moves.

Genuinely inclusive

VoicePay empowers visually impaired citizens to pay safely and independently.

Total transparency

Locked-by-default AutoPay ends hidden subscriptions and blank digital cheques.

Fair to honest sellers

Code-verified transit pool protects merchants from dishonest buyers.

Thank you · Questions welcome